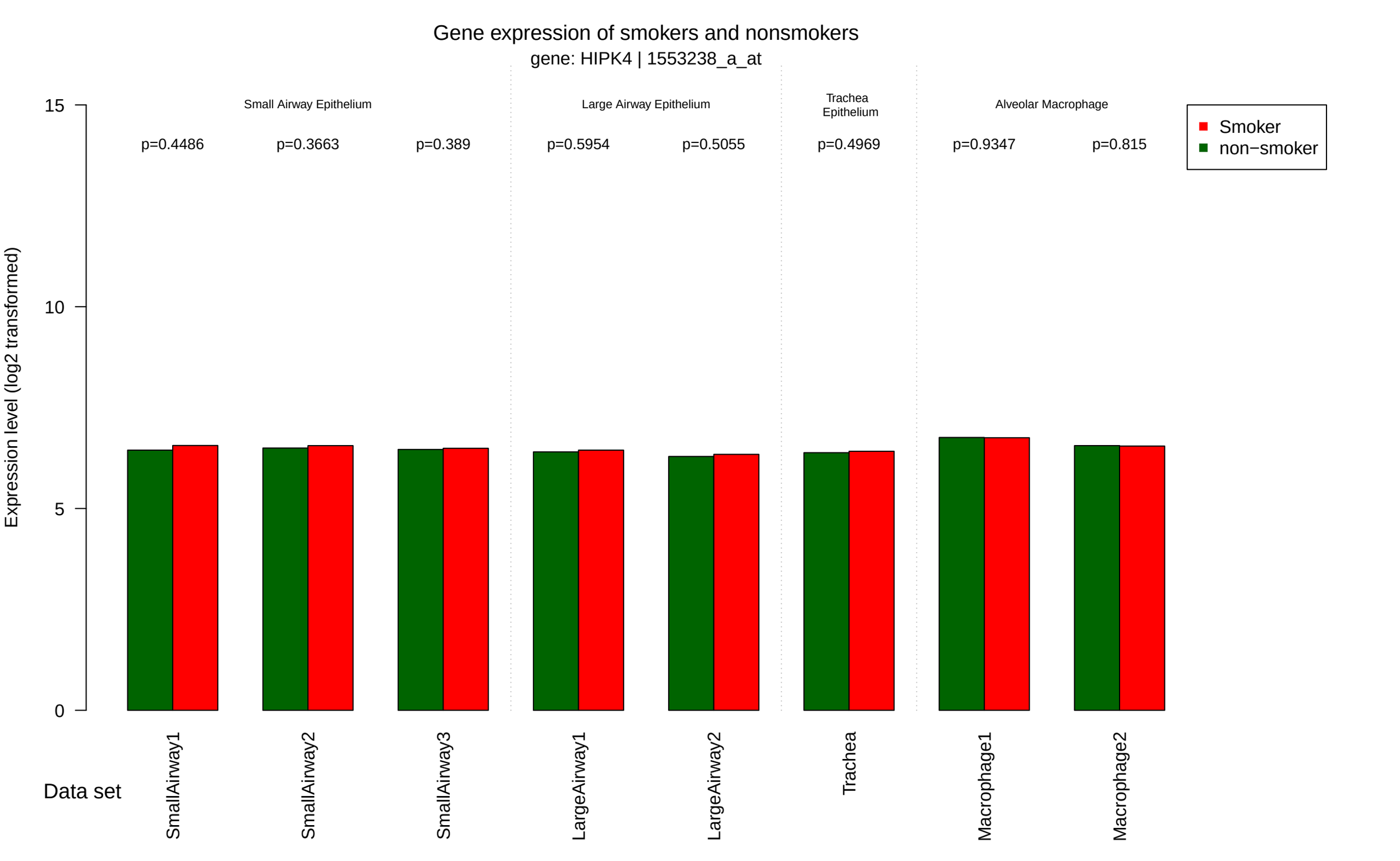
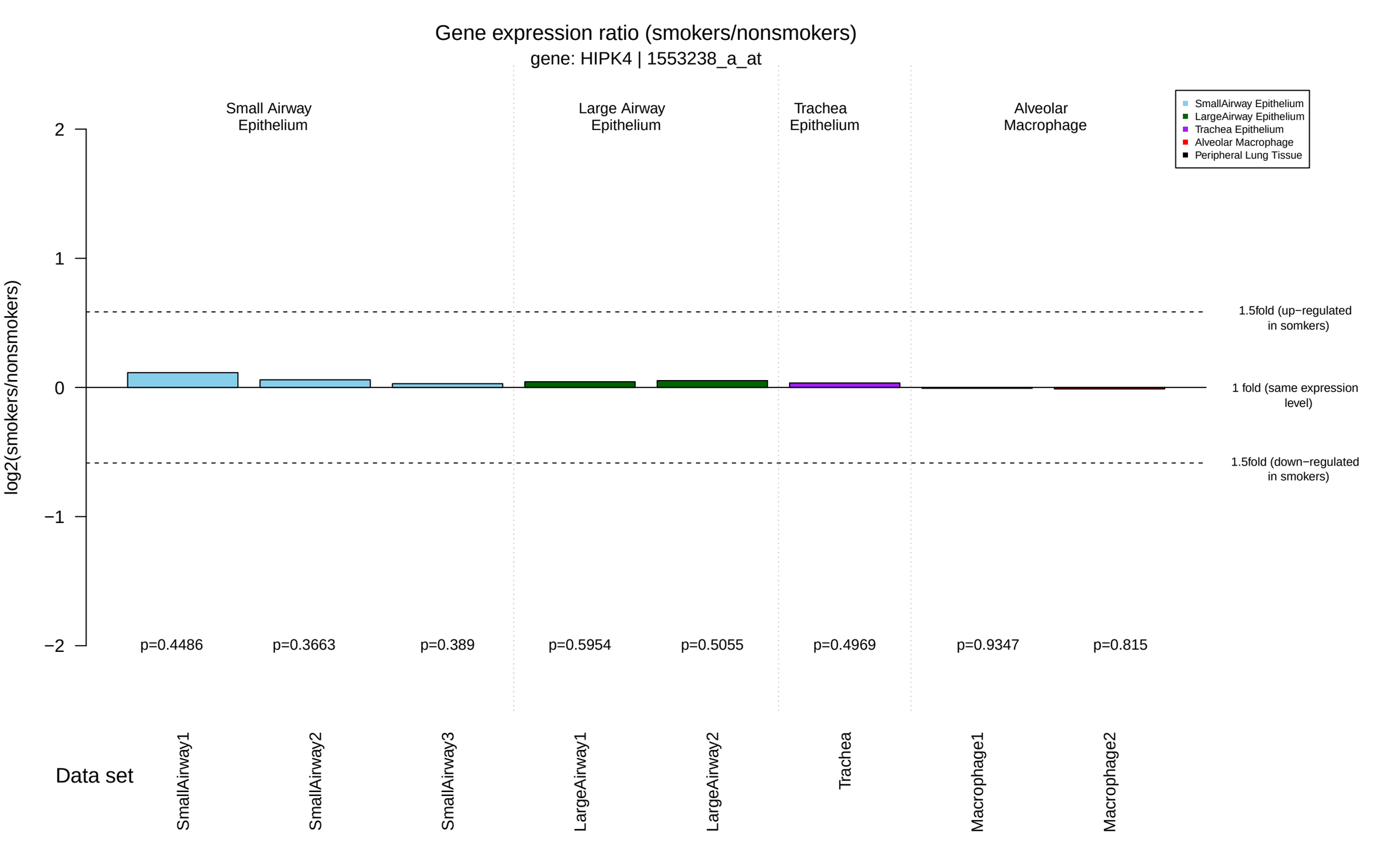
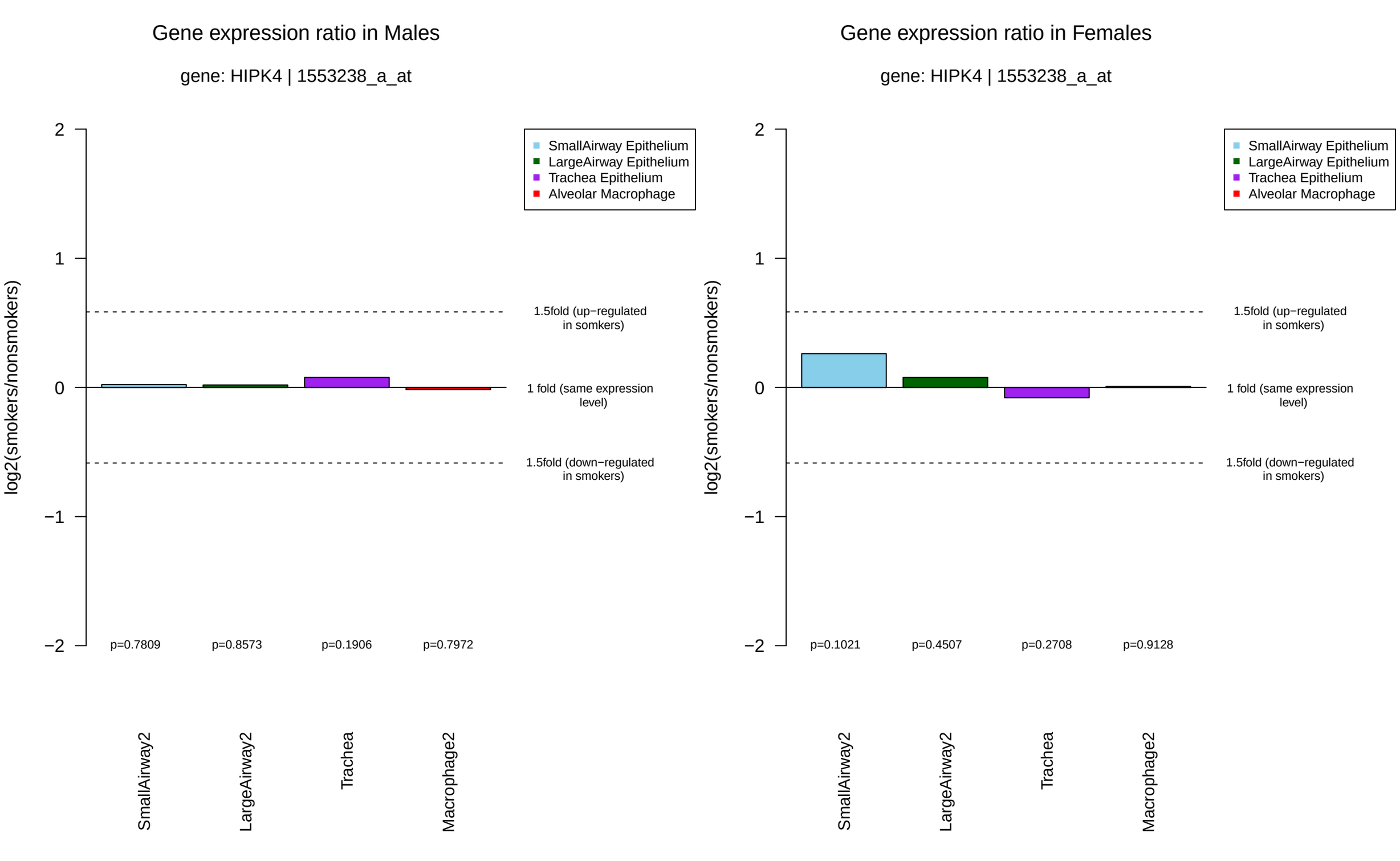
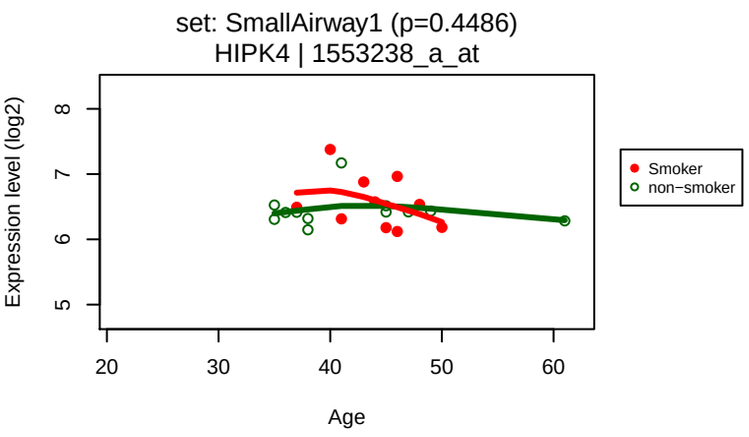
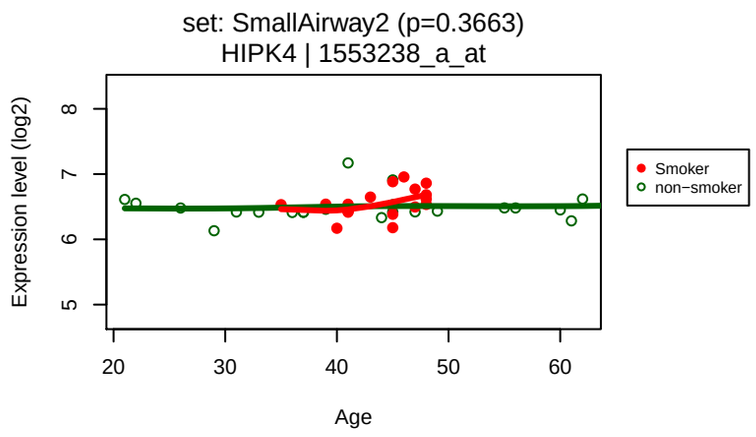
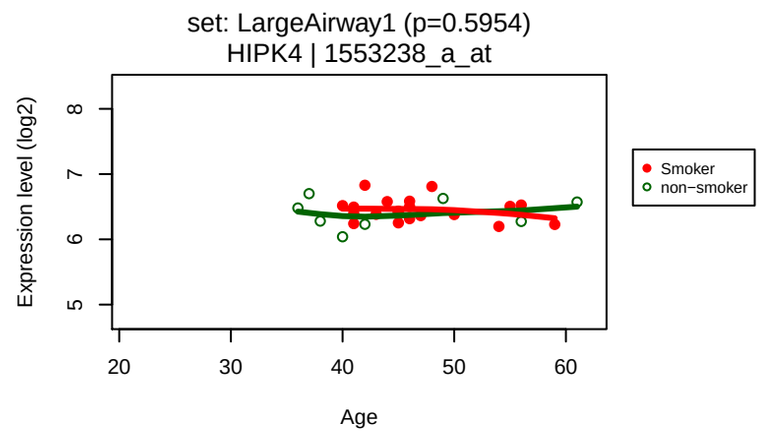

Gene expression of smokers and nonsmokers
gene: HIPK4 | 1553238_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4486
p=0.3663
p=0.389
p=0.5954
p=0.5055
p=0.4969
p=0.9347
p=0.815
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

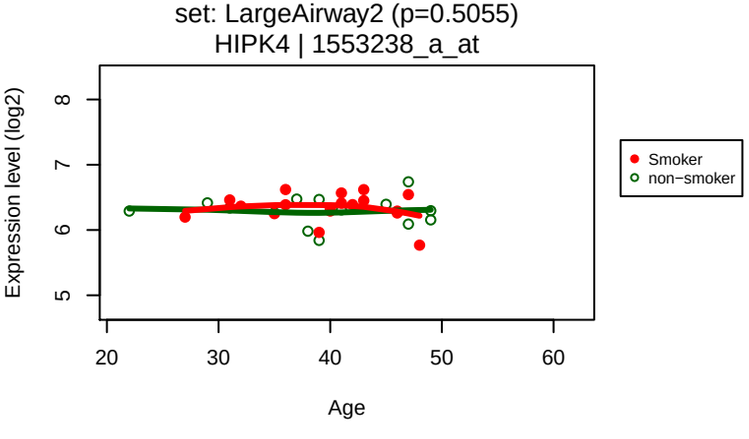

Gene expression ratio (smokers/nonsmokers)
gene: HIPK4 | 1553238_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4486
p=0.3663
p=0.389
p=0.5954
p=0.5055
p=0.4969
p=0.9347
p=0.815
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

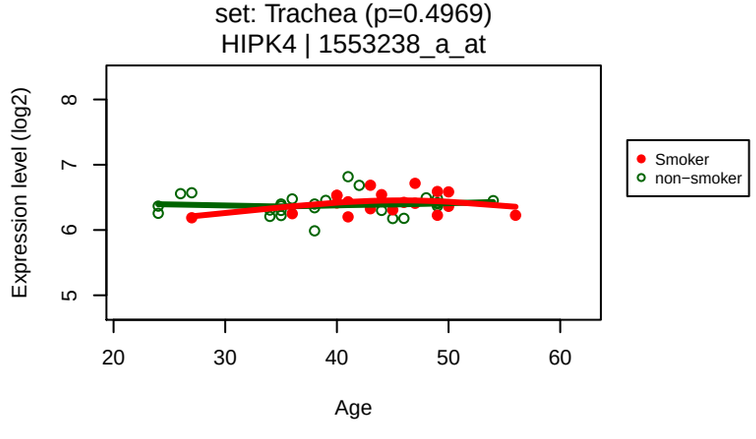

Gene expression ratio in Males
gene: HIPK4 | 1553238_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7809
p=0.8573
p=0.1906
p=0.7972
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIPK4 | 1553238_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1021
p=0.4507
p=0.2708
p=0.9128
SmallAirway2
LargeAirway2
Trachea
Macrophage2

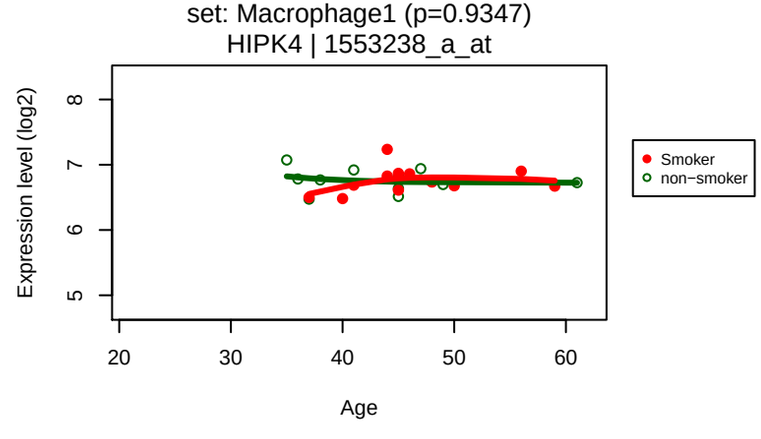

set: SmallAirway1 (p=0.4486)
HIPK4 | 1553238_a_at
Expression level (log2)
Age
Smoker
non−smoker

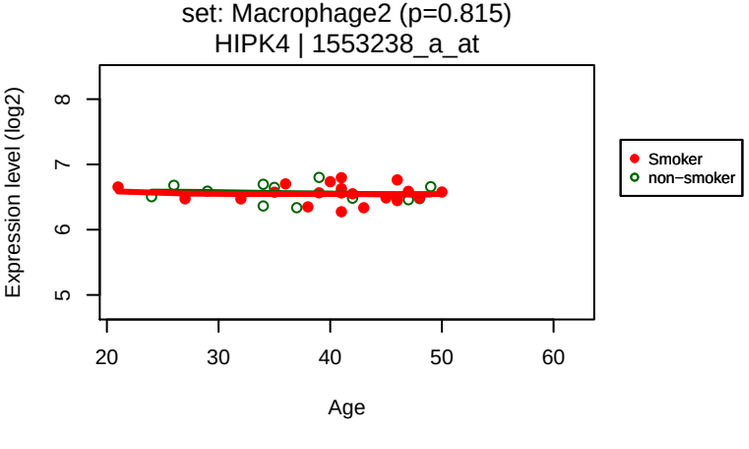

set: SmallAirway2 (p=0.3663)
HIPK4 | 1553238_a_at
Expression level (log2)
Age
Smoker
non−smoker

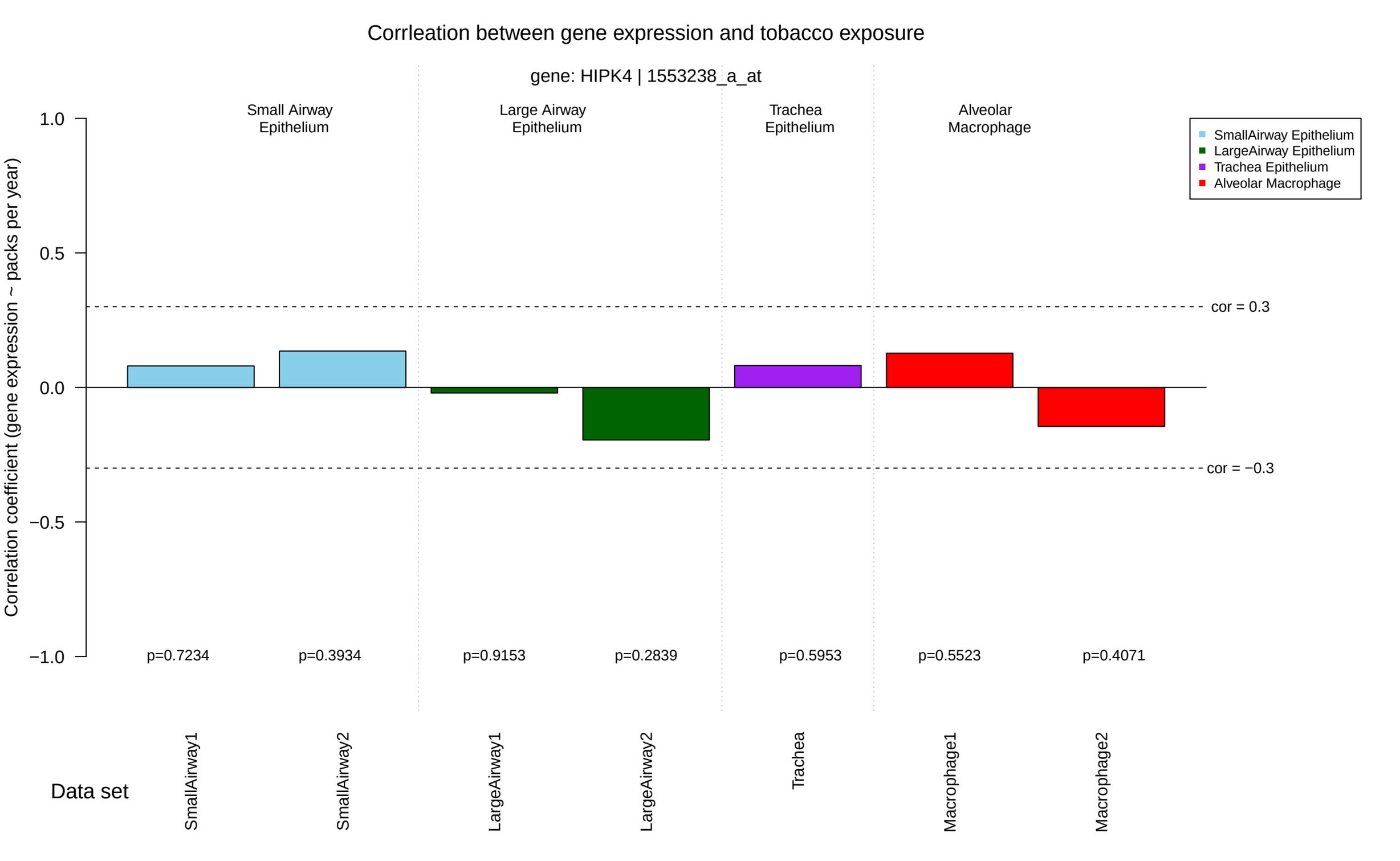

set: LargeAirway1 (p=0.5954)
HIPK4 | 1553238_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5055)
HIPK4 | 1553238_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4969)
HIPK4 | 1553238_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9347)
HIPK4 | 1553238_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.815)
HIPK4 | 1553238_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIPK4 | 1553238_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7234
p=0.3934
p=0.9153
p=0.2839
p=0.5953
p=0.5523
p=0.4071
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2